Longitudinal Embedding Drift in Facial Representations: A Study of Temporal Stability and Verification Performance

Harshinder Singh
Department of Computer Science and Engineering
ABES Institute of Technology
Ghaziabad, India

Gyan Sharma
Department of Computer Science and Engineering
ABES Institute of Technology
Ghaziabad, India

Devansh Goel
Department of Computer Science and Engineering
ABES Institute of Technology
Ghaziabad, India

Siddhant Nirwal
Department of Computer Science and Engineering
ABES Institute of Technology
Ghaziabad, India

***Abstract*—Face recognition systems typically evaluate whether two images belong to the same identity, but a different question is equally important: how stable is the numerical representation of the same person across longitudinal observations? We study this question using MORPH, a longitudinal face database, and fixed ArcFace-based embeddings. We construct an identity-disjoint train/validation/test protocol, measure embedding drift as $1 - \cos(\mathbf{e}_i, \mathbf{e}_j)$, and relate drift to genuine verification outcomes. In this MORPH cohort, the same-person face embedding was not perfectly stable across time, and higher observed drift was associated with larger longitudinal age gaps. Identity-level drift was heterogeneous, indicating that representation change is not uniform across individuals. Verification performance remained strong at the fixed operating threshold in the held-out test split, but failure rates increased markedly for larger age gaps and for high-drift comparisons. We interpret these findings as observational evidence about representation stability and verification behavior, not as causal proof of aging effects or a validated re-enrollment policy. The study motivates future drift-aware re-enrollment research while emphasizing that adaptive re-enrollment remains a direction for future experimental validation.**

***Index Terms*—Face recognition, face embeddings, longitudinal analysis, embedding drift, ArcFace.**

I. Introduction

Face recognition has become a mature research area, yet most evaluation protocols focus on matching two images at a single point in time or under controlled acquisition conditions. Longitudinal face data introduces a different problem: the same identity is observed repeatedly over time, and the face representation may drift as a function of age, acquisition conditions, and other covariates. This motivates a complementary question to the usual same-vs-different identity task: how stable is a person's face embedding across repeated observations, and how is that stability related to verification performance?

This paper studies this question in a longitudinal MORPH setting using fixed ArcFace embeddings. We distinguish between conventional recognition accuracy and representation stability. A face-recognition system may still perform well overall while the underlying representation of the same identity changes measurably over time. Our objective is therefore not to claim a universal aging mechanism or a new recognition algorithm, but to characterize longitudinal embedding drift and its relationship to genuine verification outcomes.

The research question is direct: How much does a person's face embedding change across longitudinal observations, how does this embedding drift vary across individuals and longitudinal age gaps, and how is representation drift related to face-verification performance? This distinction matters because it reframes the problem from a binary identity match question to a temporal stability question. The observed drift can be used to motivate adaptive re-enrollment, but adaptive re-enrollment itself is not experimentally validated here and remains future work.

The contributions of this study are as follows. First, we characterize longitudinal drift in a large MORPH cohort using fixed, identity-disjoint train/validation/test splits. Second, we quantify how observed drift changes across longitudinal age-gap groups. Third, we evaluate the relationship between drift and genuine verification performance under a frozen threshold. Fourth, we analyze identity-level heterogeneity in representative trajectories. Fifth, we report an exploratory individualized stability model, showing that the first Identity Stability Index formulation did not add predictive information beyond current drift. Together, these findings provide a cautious empirical basis for future drift-aware re-enrollment work rather than a validated re-enrollment policy.

II. Related Work

Face recognition systems have been transformed by deep embedding models such as FaceNet [1], CosFace [2], and ArcFace [3]. These methods learn embeddings such that semantically similar faces are represented by nearby vectors, while different identities are separated in feature space. ArcFace remains a strong reference model for fixed-feature evaluations

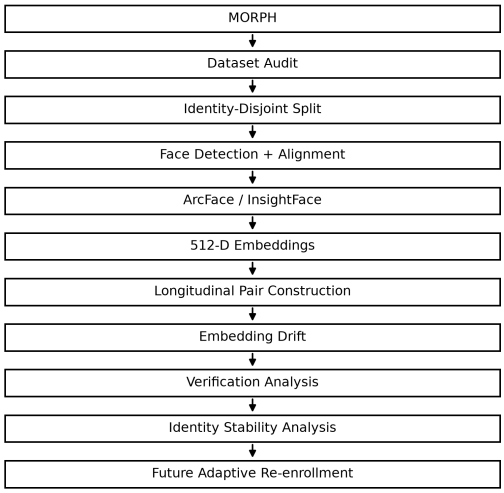


Fig. 1. Overall research pipeline. The MORPH cohort was audited, divided into identity-disjoint train, validation, and test partitions, preprocessed, converted into 512-dimensional ArcFace embeddings, and analyzed for longitudinal embedding drift and verification behavior. Adaptive re-enrollment is shown as a future research direction rather than a validated component.

because it provides a stable backbone widely used in face-recognition research.

Longitudinal face analysis is closely related to aging and template adaptation. MORPH is a classic longitudinal face dataset designed for age progression studies and repeated observations of the same identities over time [4]. AgeDB is a widely used benchmark for age-invariant face recognition and age-gap evaluation [5]. These datasets support analysis of how image appearance and model representations evolve as people age, but they do not fully isolate the causal effects of biological age from changes in lighting, pose, expression, camera, and image acquisition.

A substantial body of recognition work investigates how elapsed time and age progression affect matching performance, including studies of age-invariant recognition and template aging [6], [7]. However, much of this literature focuses on same-vs-different identity decisions rather than the representational drift of the same identity itself. Our work is intentionally different: instead of asking only whether two images match, we quantify the amount of change in the embedding of the same person over time and ask whether larger observed drift corresponds to increased verification failure.

This distinction is critical. The observed relationship between drift and failure is statistical and empirical; it does not establish causality, and it does not yet justify a validated adaptive re-enrollment system. In this sense, our study contributes to a longitudinal representation-stability perspective rather than a direct application-oriented recognition system.

III. Dataset and Experimental Protocol

A. Dataset audit and selection

We used MORPH as the primary longitudinal dataset because it contains repeated observations of the same identities across recorded ages, making it suitable for studying within-identity representation change over time. The project audit identified approximately 50,015 raw MORPH images and approximately 19,033 identities. The age variable used for modeling and ordering was the filename-derived age, which is an available longitudinal/acquisition variable but not a perfect biological-age measurement. This distinction is important: the age labels are useful for ordering and age-gap analysis, but they should not be interpreted as exact physical age or a causal treatment of aging.

AgeDB was also audited as a secondary dataset. It contains approximately 12,240 images from about 440 identities with four protocol files for 5-year, 10-year, 20-year, and 30-year splits, each with about 6,000 records and roughly 3,000 genuine and 3,000 impostor comparisons. During image-level reconstruction, 15,716 records were ambiguous or unresolved. We therefore did not use AgeDB as the primary image-level embedding validation dataset for the main longitudinal MORPH study. It is noted here briefly because it supports the broader literature context and the project audit, but it was not the main evidence source for the claims in this paper.

TABLE I
Dataset summary.

Dataset property	Value	Notes
MORPH raw images	50,015	Primary longitudinal dataset
MORPH successfully processed	50,013	2 failed cases
MORPH identities (raw audit)	19,033	Approximate audit count
AgeDB images	12,240	Secondary audit dataset
AgeDB identities	440	Approximate audit count
AgeDB protocol files	4	5-, 10-, 20-, 30-year
Eligible MORPH longitudinal pairs	84,133	Same identity, different observation

TABLE II
Identity-disjoint research split.

Split	Identities	Images	Longitudinal pairs	Notes
Train	13,323	35,040	59,545	No identity overlap
Validation	2,855	7,581	12,303	No identity overlap
Test	2,855	7,394	12,289	No identity overlap

B. Identity-disjoint research split

The original MORPH dataset was not organized for an identity-disjoint longitudinal evaluation objective. To reduce identity leakage and to provide a cleaner longitudinal analysis, we constructed a research split at random seed 42. The resulting split was identity-disjoint across train, validation, and test partitions.

The final research split contained 13,323 train identities, 2,855 validation identities, and 2,855 test identities with no identity appearing across splits. This design was chosen to reduce identity leakage and to provide a more reliable evaluation of longitudinal drift in a held-out setting.

C. Longitudinal pair construction

After preprocessing, the project produced 84,133 eligible same-identity longitudinal pairs. A pair was defined as two

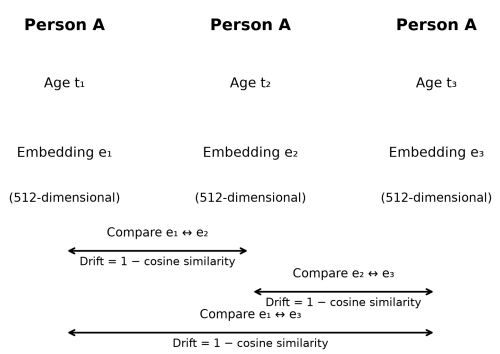


Fig. 2. Illustration of longitudinal embedding comparison. Multiple observations of the same identity at different recorded ages produce embeddings e_1, e_2, and e_3. Pairwise cosine similarity is used to compute embedding drift between longitudinal observations.

different observations of the same identity with an age gap computed as the absolute difference in filename-derived age. The analytical reference was the youngest observation by filename-derived age, with filename tie-breaking as a deterministic ordering rule. This analytical reference is not necessarily a true enrollment image; it is an analytical baseline used to study longitudinal changes in a fixed representation space.

The longitudinal coverage was substantial: 13,119 identities had at least two observations, 7,514 had at least three, 3,872 had at least four, and 2,188 had at least five. Longer-span subsets were also present, including 326 identities with at least three observations spanning at least five years, 145 spanning at least ten years, and 26 spanning at least twenty years.

IV. Methodology

A. Preprocessing and embedding extraction

Raw MORPH images were processed with InsightFace face detection and ArcFace-compatible alignment using the standard 112×112 crop procedure. The preprocessing stage handled 50,013 of 50,015 raw images successfully, with 2 failures: `138612_3F55.JPG` and `45148_00M40.JPG`. Detection outcomes were 2 zero-face cases, 50,013 single-face cases, and 0 multiple-face cases. The aligned output was stored as BGR uint8; model normalization was deferred to the embedding-extraction stage.

We used InsightFace with the `buffalo_l` model and extracted 512-dimensional ArcFace embeddings. We then applied L2 normalization. Quality checks confirmed 50,013 successful embeddings, 0 extraction failures, 0 missing embeddings, 0 unexpected embeddings, and all embeddings were finite, 512-dimensional, and approximately unit-normalized.

B. Drift definition

For normalized embeddings $\mathbf{e}_i$ and $\mathbf{e}_j$, we define cosine similarity as

$$\text{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i^T \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}. \tag{1}$$

Embedding drift is defined as

$$\text{drift} = 1 - \text{sim}(\mathbf{e}_i, \mathbf{e}_j). \tag{2}$$

Here, low drift indicates a more similar representation and high drift indicates larger representation change. Euclidean distance was retained as a secondary consistency metric, but the primary metric is cosine-based drift. We do not interpret drift itself as recognition error; it is a representation change metric.

C. Identity-level and age-gap analyses

We computed identity-level drift summaries across all available same-identity longitudinal pairs. The identity-level distribution showed substantial heterogeneity, which is relevant because some identities changed more than others even under the same model and dataset. We also grouped observations by age gap: 0–2, 3–4, 5–9, 10–19, and 20+. This analysis was descriptive and observational.

D. Verification protocol

For verification, we used genuine longitudinal comparisons and a deterministic within-split impostor set constructed with random seed 42. The primary threshold was selected on the validation split using the empirical 1% impostor FAR rule and then frozen before testing. The selected similarity threshold was 0.206770330667, corresponding to a drift boundary of 0.793229669333. This operating threshold should be interpreted as a verification boundary, not as a validated adaptive re-enrollment threshold.

V. Results

A. Identity-level drift

The identity-level summary across 13,119 identities was as follows: mean drift 0.2296, standard deviation 0.0813, median 0.2180, minimum 0.0559, maximum 1.2184, P25 0.1818, P75 0.2620, P90 0.3118, P95 0.3469, and P99 0.4653. These values show that representation change is not negligible even for the same identity and that there is substantial between-identity heterogeneity. The distribution is visualized in Fig. 3.

B. Age-gap drift

We observed increased mean drift with larger age gaps. For the MORPH pairs analyzed here, mean drift was approximately 0.236 for 0–2 years, 0.261 for 3–4 years, 0.305 for 5–9 years, 0.375 for 10–19 years, and 0.774 for 20+ years. The age-gap effect is shown in Fig. 4. This relationship was observational and consistent with the idea that larger temporal separation coincided with larger representation change in the evaluated dataset.

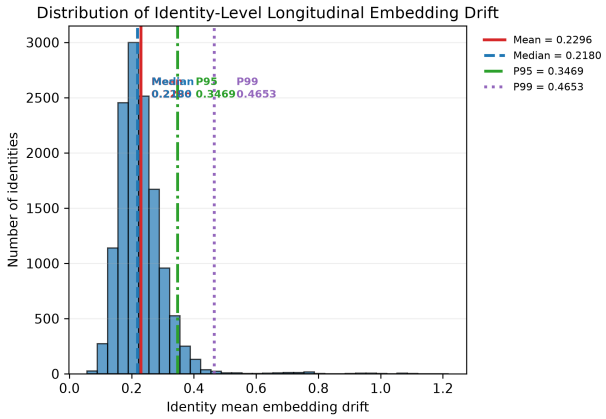


Fig. 3. Distribution of identity-level mean longitudinal embedding drift across 13,119 identities. Mean, median, P95, and P99 markers illustrate the substantial heterogeneity in observed representation change across identities.

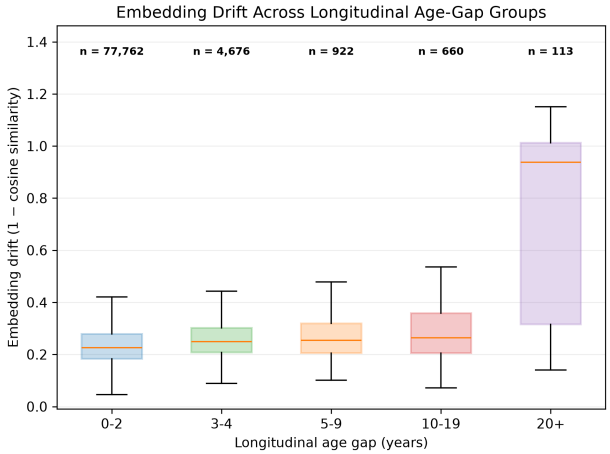


Fig. 4. Distribution of embedding drift across longitudinal age-gap groups. Observed drift increased with longitudinal separation in the evaluated MORPH sample. The 20+ year group contains relatively few observations and should therefore be interpreted cautiously.

TABLE III
EMBEDDING AND VERIFICATION PERFORMANCE.

Split	ROC-AUC	EER	FAR	FRR
Train	0.996767	0.002625	0.007598	0.005250
Validation	0.996745	0.002751	0.010157	0.005501
Test	0.996201	0.003745	0.008592	0.007270

TABLE IV
AGE-GAP DRIFT AND VERIFICATION PERFORMANCE ON THE TEST SPLIT.

Age gap	Pairs	Mean drift	Acceptance	FRR
0–2	11,114	0.2382	99.93%	0.07%
3–4	836	0.2693	99.76%	0.24%
5–9	219	0.3363	93.61%	6.39%
10–19	89	0.3662	88.76%	11.24%
20+	31	0.6681	38.71%	61.29%

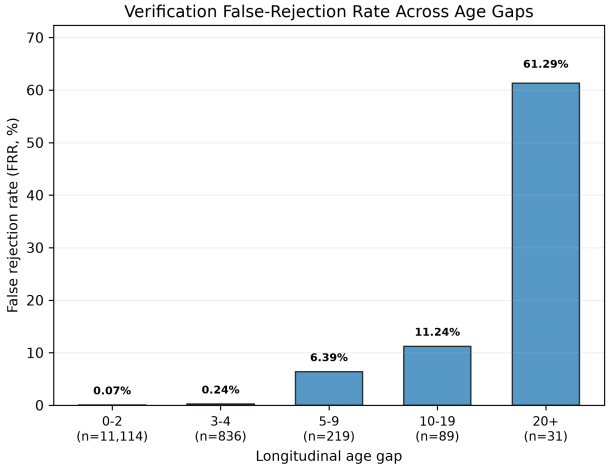


Fig. 5. Test-set false rejection rate across longitudinal age-gap groups. Verification failure increases for larger longitudinal separations in the evaluated sample. The 20+ year group contains only 31 test pairs and should not be generalized beyond the evaluated sample.

C. Verification performance

The held-out test result remained strong at the frozen verification threshold. The test split achieved ROC-AUC 0.996201, EER 0.003745, FAR 0.008592, and FRR 0.007270. The validation and train performance were comparably strong (Table III).

D. Age-gap verification failure

At the test threshold, verification failure increased with age gap. The test-set FRR was 0.07% for 0–2 years, 0.24% for 3–4 years, 6.39% for 5–9 years, 11.24% for 10–19 years, and 61.29% for 20+ years. The total pair counts were 11,114, 836, 219, 89, and 31, respectively. The 20+ group is small and should not be interpreted as a general population estimate.

E. Drift bins and operating boundary

The drift-bin analysis showed a consistent pattern: moderate drift values were still compatible with positive verification decisions, whereas sufficiently large drift values crossed the fixed verification operating boundary. The threshold at 0.7932 is shown in Fig. 6 and is interpreted as a verification boundary, not as a validated re-enrollment threshold.

F. Longitudinal trajectories

The trajectory analysis showed heterogeneous identity-specific patterns. Some identities had relatively stable drift behavior across repeated observations, others showed increasing drift with age, and some had nonlinear or irregular movement. Candidate return/persistent events were identified using a training-derived exploratory threshold of 0.361454629, but these are descriptive candidate events only and not evidence of a physiological mechanism or a production rule. Representative trajectories are shown in Fig. 7.

VI. DISCUSSION

The main empirical result is not that embedding drift is a universal failure mode, but that it is measurable and varies

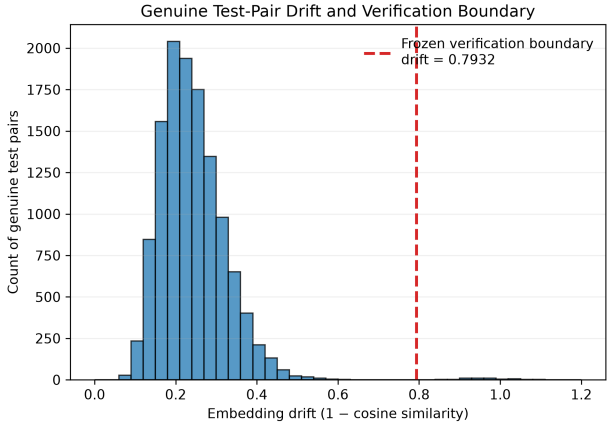


Fig. 6. Distribution of genuine test-pair embedding drift and the frozen verification operating boundary at drift = 0.793229669333. The boundary represents the selected verification operating point and should not be interpreted as a validated re-enrollment threshold.

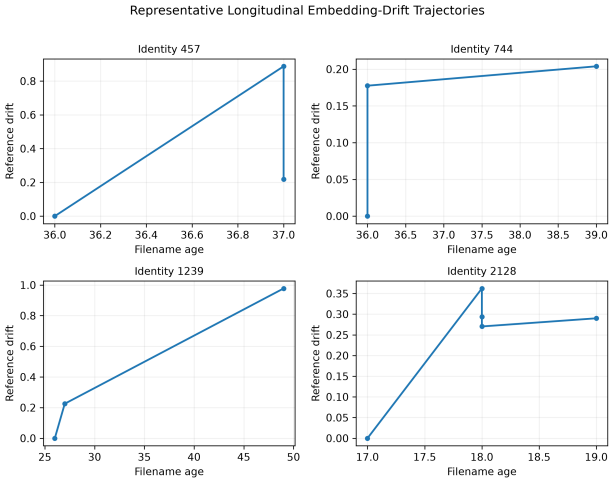


Fig. 7. Representative longitudinal embedding-drift trajectories. The examples illustrate relatively stable, increasing, and heterogeneous temporal behavior across identities. These trajectories are descriptive and do not imply a causal mechanism.

with longitudinal separation. In the evaluated MORPH data, drift increased with larger age gaps, and larger age gaps were associated with higher FRR. The effect is not uniform across identities: the distribution of identity-level mean drift was broad, and some identities remained stable while others accumulated larger representation changes over time.

This pattern matters because it clarifies the distinction between representation stability and recognition success. A fixed recognition model can remain highly accurate overall even when substantial drift is present in some observations. The fixed ArcFace representation is not perfectly static across time, but the decision boundary remains strong enough to preserve high performance in many observed cases.

The data also show that high drift does not automatically imply verification failure; moderate drift can still be compatible with successful matching under a fixed threshold. Therefore, the drift boundary should be interpreted as a behavioral characterization in drift space, not a validated re-enrollment trigger.

TABLE V
EXPLORATORY ISI COMPARISON.

Model	ROC-AUC	PR-AUC	Brier
Current reference drift	0.661054	0.367662	0.037012
ISI	0.661054	0.367662	0.037012

VII. EXPLORATORY IDENTITY STABILITY INDEX

We also evaluated an exploratory individualized Identity Stability Index (ISI). This index used current reference drift and a small set of identity-specific historical terms, including historical percentile, historical z-score, median deviation, consecutive drift, and recent slope.

The final observed result was negative for the first formulation: the corrected production artifact showed that the ISI score had ROC-AUC 0.661054 and PR-AUC 0.367662, exactly matching the current reference drift baseline. The correlation between current drift and the ISI was 1.0. Therefore, in this dataset and with this implementation, the ISI did not demonstrate independent predictive value beyond current drift.

VIII. LIMITATIONS

This study has several important limitations. First, MORPH is observational; the data capture repeated observations but not controlled longitudinal interventions. We therefore cannot infer a causal effect of age on embedding drift. Second, the age variable is filename-derived and is not a perfect biological-age measurement. Third, observed drift may reflect multiple factors in addition to age, including illumination, pose, expression, facial hair, image quality, camera/device, compression, and appearance changes. Fourth, the 20+ year age-gap group contains only 31 test pairs. Fifth, the fixed ArcFace model is pretrained and not re-optimized for this longitudinal setting. Sixth, the first ISI formulation was exploratory and did not add independent predictive value. Seventh, the Stage 8 bootstrap methodology was declared, but the executed analysis did not perform the intended bootstrap, so confidence intervals for the ISI analysis should not be claimed. Eighth, AgeDB was audited for supplemental context but was not used as the main image-level longitudinal validation set. Ninth, any re-enrollment policy remains speculative and future work is required before any adaptive re-enrollment system can be considered validated.

IX. CONCLUSION AND FUTURE WORK

This paper provides an empirical characterization of longitudinal embedding drift in a large MORPH cohort. We observed that the same person's representation was not perfectly stable across repeated observations, that drift was heterogeneous across identities, and that larger longitudinal age gaps were associated with higher observed drift and lower verification acceptance in the evaluated sample. At the fixed verification

threshold, the held-out test performance remained strong overall, but long-term separations exhibited substantially greater verification failure rates.

These findings support a temporal stability interpretation rather than a causal or policy claim. They motivate future work on drift-aware re-enrollment, but do not validate an adaptive re-enrollment algorithm. Future work should examine genuinely independent individualized stability models, identity-cluster bootstrap uncertainty, and policy simulations comparing no re-enrollment, fixed-period re-enrollment, drift-triggered re-enrollment, and individualized re-enrollment.

REFERENCES

[1] F. Schroff, D. Kalenichenko, and J. Philbin, "Facenet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 815–823.

[2] H. Wang, Y. Wang, Z. Zhou, X. Ji, D. Gong, J. Zhou, Z. Li, and W. Liu, "Cosface: Large margin cosine loss for deep face recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 5265–5274.

[3] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4690–4699.

[4] K. Ricanek and T. Tesafaye, "Morph: A longitudinal image database of normal adult age-progression," in *Proc. 7th International Conference on Automatic Face and Gesture Recognition*. IEEE, 2006, pp. 341–345.

[5] S. Moschoglou, A. Papaioannou, C. Sagonas, J. Deng, D. Kollias, and S. Zafeiriou, "Agedb: The first manually collected, in-the-wild age database," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 51–60.

[6] U. Park, Y. Tong, and A. K. Jain, "Age-invariant face recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 32, no. 5, pp. 947–954, 2010.

[7] H. Han, C. Otto, and A. K. Jain, "Age estimation from face images: Human vs. machine performance," in *Proc. International Conference on Biometrics (ICB)*, 2013, pp. 1–8.